

MCR3M - M4C3

Agenda:

0. Reminder: Complete Learning Skills Self Assessment
1. Minds-On: Summarize Quadratic Skills/Methods
2. Action: Solving Problems
3. Consolidation: GoFormative Exit Card

Learning Goal:

I will create a quadratic model to solve quadratic application problems



Learning Skills Self Assessment

If given an equation in...

Standard Form:

$$f(x) = ax^2 + bx + c$$

- direction of opening: a
- **y-intercept:** c

To find **zeroes**:

1. Factor
2. Use quadratic formula

To find **vertex**:

1. Factor, find axis of symm, use it to determine y-value of vertex
2. Complete the square

Factored Form:

$$f(x) = a(x - r)(x - s)$$

- direction of opening: a
- gives **zeroes**:
(r, 0) and (s, 0)

To find **vertex**:

1. Factor, find axis of symm, use it to determine y-value of vertex

Vertex Form:

$$f(x) = a(x - h)^2 + k$$

- direction of opening: a
- **vertex:** (h, k)

To find **zeroes**:

1. Expand, use quadratic formula
2. Rearrange to solve for x

If given a Table of Data...

Create a quadratic model: Identify key information from table (vertex, zeroes) to create either a vertex form or factored form equation template. Use any other point to find the **a** value in equation.

Ex. 1: The height of ball, $h(t)$, after t seconds, is represented by the equation $h(t) = -5(t - 4)^2 + 70$.

- a) What is the maximum height the ball reaches?
- b) How long is the ball in the air for?

Ex. 2: The population of a rural town can be modelled by the function $P(x) = 3x^2 - 102x + 25\,000$, where x is the number of years since 2000. According to the model, when will the population be lowest?

Ex. 3: The height of an arrow shot by an archer is given in the table. Determine the equation for a curve of good fit. State any restrictions on the domain and range of your model. Use it to predict when the arrow will hit the ground.

Time (s)	0	0.5	1.0	1.5	2.0	2.5
Height (m)	0.5	9.2	15.5	19.3	20.5	19.3

Ex. 4: The number of new cars sold in Canada from 1982 to 1992 is shown in the table.

Year	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
New Cars Sold (000s)	718	841	971	1135	1102	1061	1056	985	885	873	798

- a) Use Desmos to determine the equation of a curve of good fit.
- b) How well does the equation predict car sales after the year 2000? Explain.

Consolidation:

Complete GoFormative M4C3

